EcoWatt AI

Household energy awareness and actionable saving scenarios
Samsung Innovation Campus | Group 8 | Project Report | 17 October 2026

01 Problem, users and SDG alignment

A household electricity bill shows a total, but does not explain which appliances and habits contribute to it. People without electrical knowledge may also be unable to provide device ratings. EcoWatt AI serves household residents who want an accessible estimate and practical ways to review their electricity use. Its primary alignment is **SDG 7, target 7.3: improving energy efficiency** [1]. It also supports **SDG 12.8** by promoting awareness of sustainable consumption [2]. In addition, it has a potential connection to **SDG 13: Climate Action**, particularly the awareness purpose of target 13.3 [3]. Sustained electricity savings could help reduce emissions associated with electricity generation. These are contribution pathways, not evidence that the prototype has already achieved energy savings or emission reductions at household or national scale.

02 Solution and model

The English-language Python and Streamlit application [4] collects household context, an optional past electricity bill in kWh, and appliance groups with quantities. Users may provide known values or retain editable, explicitly labelled reference estimates. Appliance-specific inputs include W and hours/day, kWh/year, kWh/cycle or kWh/100 cycles with cycles/week, and phone battery Wh with charging frequency and efficiency. Home type and resident count do not apply an arbitrary multiplier.

The integration uses Google Gemini through the google-genai SDK. The repository configures **gemini-3.8-flash** by default; GEMINI_MODEL can override the configured identifier. Python calculates consumption and savings. Gemini receives compact household/device JSON, energy shares and input provenance, then returns a short analysis and up to three proposed changes. This division keeps model-generated advice separate from deterministic arithmetic.

03 Calculation and interaction design

Power-based energy is W / 1,000 x hours/day x days; annual energy is prorated by days / 365; weekly cycles by days / 7. Phone energy includes charging losses. Quantity is applied once. Calculations retain full precision; values are rounded for display and the AI input summary. A proposed change is evaluated on a copy of the device by **potential_saving()**: current consumption minus scenario consumption. Original inputs remain unchanged.

Users explicitly approve data sharing and press Analyze with Gemini. Results include an overview, device-specific action cards and Python-calculated potential savings. A fingerprint marks the analysis Out of date after inputs change. Combined savings are shown only when suggestions concern different device groups, avoiding overlap.

04 Demonstrated outcome and limits

In a team-designed manual test scenario, a 30-day inventory for four residents, including three phones, one TV and other household appliances, produced **142.3 kWh**. During a manual test, the team obtained a live response proposing changes for a desktop computer, dishwasher and lighting; Python calculated a combined **17.3 kWh/month** potential saving. This demonstrates the workflow, not measured savings. The entered 200 kWh past bill was a test baseline; the 57.7 kWh gap does not establish predictive accuracy. The project README reports 21 passing tests with fake-client Gemini coverage; the team also reported a successful manual end-to-end test. No longitudinal household study has been conducted. The demonstration video is submitted separately and may use a different scenario.

In this example, the proposed reductions are desktop use from 5 to 3.5 hours/day, dishwasher use from 5 to 4 cycles/week, and lighting from 5 to 3.5 hours/day. Together they represent approximately **12% of the estimated total**. This scenario assumes the changes are practical and maintained throughout the same 30-day period; it does not establish that the current usage is wasteful. Users can review the assumptions before deciding what to change. The difference from the past bill is a separate comparison: it is neither a saving nor proof of missing devices. This distinction helps users understand what the application has calculated and what still needs to be checked against real household conditions.

Prompt design & responsible use

Explicit constraints, validated scenarios and transparent uncertainty

05 Most important prompts and rationale

Prompt version 1.0 is stored in **prompts.py**. Its design rationale is documented in **PROMPTS.md**. The instructions define the assistant role, allowed data, units, safety rules and English tone. Key excerpts follow:

> “Use only the data provided. Never invent appliances, measurements, prices, tariffs or brand names.”
>
> “Do not calculate or state kWh savings yourself. The application calculates savings from the value you propose.”
>
> “Inputs with source "reference_estimate" are assumptions, not user data.”

Task decomposition ranks consumers, checks provenance and selects a realistic change, returning only final JSON. It does not guarantee correctness. A **one-shot example** illustrates a heater change from 3 to 2 hours/day, acknowledges estimated usage and anchors tone and format.

A JSON schema restricts device IDs and fields. Server-side validation checks the device-field pairing and value, rejecting increases or zero-consumption scenarios. Prompt instructions alone are insufficient.

06 Challenges and remaining limitations

Units and appliance behaviour: maximum nameplate power is not always average input, particularly for thermostatic or cycling equipment. Appropriate input modes and explanatory labels reduce this risk.
Incomplete inventories and uncertain habits: missing appliances, standby loads, seasonality and generic reference values can create bill differences. The app exposes assumptions and does not force its estimate to match a bill.
Advice realism: arithmetic validation cannot establish whether a household can comfortably reduce usage or combine dishwasher loads. Advice must be conditional and should not promise that a particular reduction meets every household need.
Reliability: the API layer handles key, quota, network and malformed-response errors. Repeated analyses can vary. Automated tests and one live demo do not prove accuracy across all homes or every failure condition.

07 Ethics, privacy and impact claims

Consumption, recommendations and potential savings are estimates, not meter readings or guarantees. Numerical correctness is distinct from the accuracy of assumptions. Safety instructions prohibit unsafe refrigerator use, electrical modifications and advice that compromises necessary heating or cooling. These are mitigations, not guarantees of safe AI output.

No name or address is requested. Household figures, device names, usage and provenance are sent to Google only after consent and an explicit action. Users are warned against personal information in labels. API keys remain in session/local configuration and must not be committed. User-written names are treated as labels; prompt rules and schema restrictions reduce, but do not eliminate, injection risks. Provider data handling still applies.

Environmental benefits require sustained adoption. Collective scenarios need explicit assumptions; carbon claims need regional, dated emission factors. Next steps are user testing and same-period bill follow-up accounting for weather and occupancy.

Sources and implementation evidence

[1] United Nations, SDG 7 (target 7.3): sdgs.un.org/goals/goal7

[2] United Nations, SDG 12 (target 12.8): sdgs.un.org/goals/goal12

[3] United Nations, Goal 13: Take urgent action to combat climate change and its impacts (target 13.3): sdgs.un.org/goals/goal13

[4] Project: github.com/yagiz-boz/SIC-Hackathon-Group-8. Reviewed local files: core.py, ai.py, prompts.py and PROMPTS.md; README.